Primary 3 · Science

Science Open-Ended Drill · Set 1 — Answer Key

24 marks · 25 min

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Quick Drill: Explaining Science Answers **(24 marks)**

No.	Answer	Marks
1	Inside the pupa, the caterpillar's body changes completely into a butterfly. It does not eat or move around during this stage — it stays still while its body changes shape. — Marking guide: 1 mark for describing a complete body change, 1 mark for noting it does not feed/move during this stage, 1 mark for a clear overall explanation. | Hint: Say what happens to its BODY, and whether it eats or moves during this stage.	3
2	A tadpole goes through big body changes as it grows — it loses its tail, grows legs, and its gills are replaced by lungs. A chick already has a body shape similar to a hen and mostly just grows bigger, without such big changes. — Marking guide: 1 mark for describing a tadpole's body changes, 1 mark for noting the chick does not change shape much, 1 mark for a clear comparison. | Hint: Compare BOTH animals: say what happens to one, then what happens to the other.	3
3	Mushrooms are fungi, not plants. Unlike green plants, which make their own food using sunlight, fungi cannot make their own food — they feed on dead or decaying matter instead. — Marking guide: 1 mark for stating mushrooms are fungi, not plants, 1 mark for stating fungi cannot make their own food (unlike plants), 1 mark for a clear overall explanation. | Hint: Say what GROUP mushrooms belong to, and how they get their FOOD.	3
4	Even though both reproduce using spores, ferns are plants that can make their own food using sunlight, while mushrooms are fungi that cannot make their own food — they feed on dead or decaying matter instead. This is the key difference between them. — Marking guide: 1 mark for noting spores alone do not make them the same, 1 mark for stating the food-making difference (plants use sunlight, fungi cannot), 1 mark for a clear overall explanation. | Hint: Spores are not the key difference — think about how each one gets its FOOD.	3

5	Only the type of ball was changed. The height dropped from and the floor were kept the same for every ball. This makes it a fair test because any difference in how high the balls bounce is caused only by the type of ball, not by any other factor. — Marking guide: 1 mark for stating what was changed (the ball), 1 mark for stating what was kept the same (height, floor), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___.	3
6	Only the amount of sunlight was changed. The type of soil and the amount of water were kept the same for every seedling. This makes it a fair test because any difference in how the seedlings grow is caused only by the amount of sunlight, not by any other factor. — Marking guide: 1 mark for stating what was changed (sunlight), 1 mark for stating what was kept the same (soil, water), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___.	3
7	Plastic is waterproof, so rain does not soak through and wet the person wearing it. It is also flexible, so it is comfortable to move around in. These properties make it suitable because a raincoat needs to keep a person dry and be comfortable to wear. — Marking guide: 1 mark for a correct property (e.g. waterproof), 1 mark for a second correct property with its reason (e.g. flexible, so comfortable), 1 mark for a clear overall explanation linking properties to purpose. | Hint: Use the frame: ___ is suitable because it is ___, so ___.	3
8	Plastic does not break easily (it is not brittle like glass), so it is safer for a young child to use. It is also light, so it is easy for a small child to hold. These properties make it suitable because a cup for a young child needs to be safe and easy to hold. — Marking guide: 1 mark for a correct property (e.g. does not break easily, unlike glass), 1 mark for a second correct property with its reason (e.g. light, so easy to hold), 1 mark for a clear overall explanation linking properties to purpose. | Hint: Use the frame: ___ is suitable because it is ___, so ___.	3